The Spread of Misinformation on Social and Video Platforms

Race, Gender & Class in the Digital World

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Research Problem

The core research problem is the rapid, widespread dissemination of misinformation (false information spread unintentionally) and disinformation (false information spread with the intent to deceive) through digital platforms. Social media (like Facebook and X) and video platforms (like YouTube and TikTok) are the main focus of this problem. Their technological design, which is driven by algorithms, prioritizes engagement, a lack of fact-checking, and the use of automated "bot" accounts that create an environment for false content to spread faster and farther than verified, factual information. This is a critical issue because it has severe, real-world consequences. Public Health: It directly affects public health decisions, leading to vaccine hesitancy, rejection of public safety measures (such as masking), and use of unproven treatments, all of which can increase illness and mortality. Social Cohesion: It hurts public trust in institutions, including science, government, and journalism. Political stability: It divides society by strengthening echo chambers, threatening democratic processes, and, in certain situations, causing tensions and actual violence.

The immediate and significant societal impact of this problem requires rapid research; for instance, the World Economic Forum has recognized disinformation as a major worldwide danger. Understanding the psychological, sociological, and technological factors that make people and platforms vulnerable to false information needs research. By understanding why and how it spreads, researchers can develop and test effective interventions—such as platform design changes, user education, and improved fact-checking strategies to mitigate its harmful effects. This topic is a quintessential sociotechnological issue.

Technological: The problem is directly linked to the technology that makes it effective. The use of AI to produce convincing "deepfakes," the use of automated bot networks to give the appearance of widespread approval, and engagement-based algorithms that amplify emotionally charged content regardless of its truthfulness are important technological elements. The technology takes advantage of the social and cognitive tendencies of individuals. It leverages confirmation bias (our tendency to believe information that fits our existing views) and triggers strong emotional responses (like anger and fear), which make content more likely to be shared.

The social issues that result in eroding trust, political polarization, and public health crises are a direct consequence of this technological exploitation.

Literature Review

Parameters

To better understand the spread of misinformation on social media and video platforms, the literature reviewed in this study will examine several parameters. First, we examine how social media platforms and their algorithms function, particularly engagement-based systems that affect the spread of misinformation. Next, we examine the type of content being shared, including how emotional and credible the information appears to be. We also focus on user factors such as digital literacy, age, and sharing behavior to explain why some people are more likely to spread false information than others. This results in considering the impact of misinformation on public trust, beliefs, and false narratives across platforms.

Article 1 (Ariya Briscoe)

The article “Spread of Misinformation on social media: What contributes to it and how to combat it”, by Chen et al, provides an overview of research on disinformation to date. The author notes that the rise of social media has drastically changed how information spreads, making it faster, more emotional, and less verifiable than traditional media outlets. Platforms like Facebook, X, and Instagram now serve as primary sources for millions, but their engagement-driven algorithms amplify content that provokes strong reactions, regardless of its accuracy. The authors divide the variables influencing the spread of false information into four categories based on their analysis of 423 published studies. They are as follows: the source, the message, the context, and the receiver. According to their research, social media sites' algorithms, which encourage engagement and emotional content, increase the spread of misleading information among networks. Additionally, they examine the five main approaches to countering misinformation: policy development, network based interventions, message correction, enhancing source credibility, and user education. Furthermore, Chen et al. argue that stopping misinformation requires both technological advancements and human focused approaches, like better content verification systems and digital literacy training. This shows how the technical design of social media platforms, rather than just user behavior alone, plays a major role in the spread of misleading information, which is directly related to our research issue.

Article 2 (Evan Barnes)

The article “Creating a Cost to Spread Misinformation on Social Media”, written by authors Drew B. Margolin and Yunyun S. Wang, covers a study about systemic incentives that encourage the spread of misinformation on social media. What the study found was that the main reason people post harmful content on social media is due to two major factors. The first is the perpetrator's own gain in the matter, and the second is how easy it is to commit such acts online. When online, it is very simple, quick, and cheap to post anything. So people's strategy is to flood the system with everything and anything, knowing that eventually it will benefit them one way or another. On top of this, the research also found that the problem is further inhibited by a conceptual blurring between the “right to speak” and the ability to “spread”. The difference between these two is to express oneself freely and to have one's speech heard by the public at large. The main argument from this is that any move to bring consequences to these acts is met with criticism about potential violations of the First Amendment. One possible solution to this problem that was brought up by the authors was that of attestation. As written in the article, “Attestation draws a bright line between speech and spread and allows speakers autonomy in choosing the intent for their speech and thus the responsibilities they undertake for it”. This suggests that users can and should take responsibility for what they say online. It separates the act of saying something from the act of spreading it, giving users control over their intent and the responsibility that comes with it. In addition to this, the authors also point out that even though social media platforms have lots of data and power, they can’t fully understand every post or message. So, because users themselves know more about what they mean when they post something, it would be fair to blame the people who use the platforms rather than the platforms themselves.

Article 3 (Agam Aujla)

The article “The spread of true and false news online” by Vosoughi, Roy, and Aral (2018) provides a critical quantitative foundation for our research. In one of the largest-ever studies of its kind, the authors analyzed 126,000 rumor cascades shared on Twitter (now X) from 2006 to 2017. Their primary finding was that falsehood diffuses "significantly farther, faster, deeper, and more broadly" than the truth. False news was found to be 70% more likely to be retweeted than trustworthy news. Critically, the study discovered that this effect was not driven by automated "bot" accounts, which spread true and false news at roughly the same rate. Instead, the problem was driven by human users. The authors suggest this is because false news is often more "novel" and produces stronger emotional reactions, such as surprise and disgust, compared to true stories. This directly links to our research question by suggesting that human user behavior, driven by a psychological preference for novelty and emotion, is a primary engine of misinformation spread, which is then amplified by the technological design of the platforms.

Synthesis

The articles share the view that the incentives and structures of these social media and digital platforms are at the root of the systemic issue of the unchecked spread of misleading information. In their evaluation of 423 studies, Chen et al. (2022) identified four key factors that affect the spread of misinformation: source, message, context, and recipient. Additionally, they wrote a summary of the five primary methods used for countering misinformation spreading, namely policy development, user education, message correction, and increased source reliability. Their results show how social media's engagement-based algorithms and emotionally charged content spread false information faster than factual news.

One key topic for Margolin and Wang (2025) is the incentive structure that encourages "propagation hunting," where users abuse the low cost and high reward of spreading viral but misleading material. A common theme in social media algorithms is to extend the reach of these misleading posts. However, to combat this, they propose the "attestation framework," which requires users to confirm the legitimacy of their posts before they are shared, thus increasing accountability. Together, these results demonstrate that addressing misinformation requires structural and behavioral change via algorithmic transparency, user accountability, and education to change the tone on social media from one that supports misinformation to one that values legitimacy.

Research Question

How can algorithmic design, digital literacy, and user behavior interact to affect the spread of false information on social media and video platforms?

Research Design

Our team is researching how misinformation spreads on social and video platforms to understand why false information moves so quickly online and how its impact might be reduced. This issue involves both technology and society, affecting how people see truth, make choices, and use social media. Platforms like TikTok, YouTube, Instagram, and X use algorithms designed to boost engagement numbers. These algorithms do that with content that can bring out strong emotions, especially anger, fear, or surprise. Since these posts are more likely to be shared, this increases activity but also spreads misinformation. Other factors, like digital literacy and beliefs, influence how they respond to information. Together, technology and social factors create a cycle where misinformation spreads faster than truth, and many users do not realize how much platforms can alter their beliefs.

To study this problem, we reviewed academic and peer-reviewed articles that analyzed how technology, digital literacy, and user behavior affect the spread of misinformation. We will look at statistics from studies that show how algorithms recommend and boost certain content. These studies use data from social media platforms and reveal which posts are most likely to go viral and why. We selected these sources because they clearly demonstrate how technology reveals the problems, allowing us to measure the algorithm's effects on misinformation. Second, we will examine behavioral research on how people respond to emotional or misleading posts and how digital literacy affects their ability to identify false information. These articles will help us understand why people still share false posts. Finally, look into policy and communication research examining possible solutions, such as digital literacy programs, improved fact-checking, and specific guidelines for algorithms to avoid misinformation spread.

When we analyze these documents, we will look for evidence that algorithms amplify more emotionally focused or controversial content, as shown in studies that measure how social media engagement systems function. We will also examine how user behavior, such as education and digital skills, affects the likelihood of believing or sharing misinformation.

By examining both technical and social factors, we will attempt to understand how misinformation spreads and how it can be controlled. We want to learn not just how algorithms make the problem worse, but also how users’ actions and education can help address it. We will seek evidence, such as engagement metrics, user surveys, and examples of platform behavioral patterns.

In the end, we will use ideas from both technology and sociology to get a complete view of this issue. By comparing studies, we want to understand how misinformation spreads and what can be done to stop it. Through this analysis, we hope to find solutions to stop the rampant spread of misinformation.

Conduct Research

Article 1 (Ariya Briscoe)

For the first document analysis, I reviewed the article “Fake news, disinformation and misinformation in social media” by Aïmeur et al. (2023) This article provided multiple issues surrounding false information on online social networks, explaining why it spreads so quickly across platforms such as Facebook, TikTok, YouTube, and X. The authors argue that social media environments are more vulnerable to this false information because they are built with real time sharing, low barriers for posting and algorithms that are engagement driven, which means that they reward more emotional and controversial content, regardless of whether it is true or false, allowing more deceptive content to circulate. As a result, misinformation circulates faster than the fact checking systems can keep up with, reaching a larger audience.

The article also includes studies that point out important statistics that highlight the issues. For example, Aïmeur et al. report that half of American adults now get their news from social media, showing how dependent people have become on algorithm-curated news content designed to elicit emotional responses rather than on traditional sources. The author also referenced another study showing that false news spreads six times faster than truthful news, and that 70% of users cannot reliably tell the difference between fake and real content. These findings, when combined, reveal how the design of social media platforms and human behavior correlate more directly, driven by low digital literacy and emotional reactions that can lead to misinformation. Additionally, the authors explain that misinformation does not spread only on one platform but across multiple platforms. In their 2023 study, they found that False information is more likely to appear across platforms (18%) than true information (11%). This demonstrates how this false information can more easily spread through other networks due to the algorithm's recommendations and increased engagement.

Overall, this article is key to research because it clearly shows how algorithmic design, user behavior, and digital literacy interact to spread misinformation online. It directly connects to my research project by highlighting that misinformation isn’t the result of social or technological problems alone, but of a combination of both. The author's article helps explain why social media platforms' architecture and user awareness need to be addressed to reduce the spread of misinformation.

Article 2 (Evan Barnes)

For the second document analysis, I reviewed the article "The diffusion process of product-harm misinformation on social media: evidence from consumers and insights from communication professionals" by authors Chen Zifei Fay and Cheng Yang. The article investigates how product-harm misinformation spreads on social media and how it affects consumer perceptions of a company. For clarity purposes, product-harm refers to any situation where a product is believed to cause danger, damage, or negative consequences for consumers. Using a study the authors created, through a survey of 504 U.S. consumers, followed by interviews with 11 communication professionals, showed that consumers who are more skeptical of what they see online are less likely to believe false information. But if the misinformation seems believable or appears trustworthy, people are more likely to think it is true. And once consumers are convinced by said misinformation, it will damage their view of the related company, ultimately reducing their trust and increasing their likelihood of spreading negative and possibly further misinformation.

With this study and its generated information, the authors identify several strategies that communication professionals can use to reduce the spread and impact of misinformation (or specifically product-harm misinformation at the very least). To start, they share that companies should be prepared with response plans before misinformation appears. Secondly, they can actively monitor online conversations to detect false claims early. And lastly, they can work to establish a reputation for competence and reliability so that consumers are less likely to believe misleading claims to begin with.

Regarding our research question, this article further shows how digital literacy and user behavior work together to aid the growth and spread of misinformation. Furthermore, it shows how misinformation is a serious issue that can cause harm on multiple levels regarding whatever topic it is related to. And using algorithmic design, companies as well as individuals can take the author's provided steps to help mitigate the creation and spread of misinformation.

Article 3 (Agam Aujla)

For the third document analysis, I reviewed the article “Less than you think: Prevalence and predictors of fake news dissemination on Facebook” by Guess, Nagler, and Tucker (2019). This study provides specific, quantitative data on user behavior, which is central to our research question. The authors linked survey responses from a sample of Americans with their actual Facebook sharing activity during the 2016 U.S. presidential election. Their findings provide crucial context. First, they found that sharing "fake news" was a relatively rare behavior; only 8.5% of users in their sample shared at least one link from a fake news domain. Second, this behavior was highly concentrated among a specific group. The single strongest predictor of sharing misinformation was age. The study found that users over 65 shared nearly seven times as many articles from fake news domains as the youngest age group (18-29). This "age effect" remained significant even after controlling for other factors like political ideology, education, and overall digital media use.

This article is vital for our research because it directly addresses the "digital literacy" and "user behavior" components of our question with concrete statistics (8.5%, 7x, over 65). It suggests that the problem of misinformation spread may not be a widespread behavior but rather a concentrated one. The strong correlation with age points to a potential link between digital literacy (or the lack thereof in older demographics) and susceptibility to sharing false content. This provides specific, analyzable data for our project, demonstrating which user behaviors are most impactful in the spread of misinformation.

Findings

Our research problem focused on the rapid spread of misinformation and disinformation on social media and video platforms like Facebook, X, YouTube, TikTok, and Instagram. How false information spreads isn't an error, it's a social and technological problem rooted in how we interpret and understand the technology. Through our research, we reviewed several articles to examine how algorithm design, digital literacy, and user behavior interact to add to this problem.

The article by Aïmeur et al. (2023), "Fake news, disinformation and misinformation in social media," explains why social media is vulnerable to false information. The author described how the platforms are built with live data sharing, ease of access for posting, and engagement-based algorithms that benefit from emotional and controversial content, regardless of whether the content is false. They pointed out that false news spreads six times faster than true news and that many users have difficulty telling the difference between true and false. This article shows how these platforms' designs, coupled with low digital literacy, can lead to a much easier spread of misinformation, which travels faster than the fact-checking systems that are in place.

The following article we looked at was “The diffusion process of product-harm misinformation on social media: evidence from consumers and insights from communication professionals” by authors Chen Zifei Fay and Cheng Yang (2023). This research article delved into how misinformation, specifically in the realm of product harm, spreads online and alters the opinions and attitudes of consumers towards the misinformation's targeted company. The authors found that consumers who are more skeptical of online information are less likely to believe false or fabricated information. However, misinformation that appears to be credible is far more persuasive to consumers and may be accepted by them as truth. Once received, this information damages the credibility of the affected companies and increases the likelihood that the same information will be spread through word of mouth. In addition to these findings, the authors also found some key strategies for companies to take to help mitigate any potential damage. These strategies include preparing response plans, monitoring relevant online conversations, and generally fostering a positive and trustworthy reputation with consumers.

In addition to the analysis of platform structures and consumer trust, our review of the quantitative data reveals critical insights into the specific mechanics of how misinformation travels and who is spreading it. The study by Vosoughi, Roy, and Aral (2018) challenges the common assumption that automated "bots" are the primary drivers of false news. Their findings indicate that falsehood diffuses "significantly farther, faster, deeper, and more broadly" than the truth, primarily due to human action. Specifically, they discovered that false news is 70% more likely to be retweeted because it is more "novel" and evokes stronger emotional reactions, such as surprise and disgust, than true news. This finding suggests that algorithms are not working in isolation; instead, they amplify misinformation by exploiting a specific human cognitive bias toward novelty.

Furthermore, regarding the role of digital literacy, the study by Guess, Nagler, and Tucker (2019) provides a crucial demographic distinction that general platform analysis misses. Their data revealed that sharing fake news is not a universal behavior but is highly concentrated, with only 8.5% of users in their sample sharing at least one false article. Most significantly, they identified a stark generational divide: users over 65 were nearly seven times more likely to share articles from fake news domains than the youngest demographic (18-29). This quantitative evidence refines our finding on "user behavior," suggesting that the lack of digital literacy is not uniform across the population but is particularly acute among older users, who may struggle to distinguish legitimate sources within the complex architecture of social platforms.

Across these articles, multiple similar themes emerged. Both articles found that misinformation spreads further because platform design and user interactions amplify it. The researchers Aïmeur et al. (2023) highlight how algorithms reward more emotional content, pushing the reach of the content farther over the network, while Chen and Cheng (2023) explained how users accept and spread false information when it appears to be from a credible source or seems believable to their expectations. These articles emphasized that low digital literacy plays a big role in how fast misinformation can be shared. Additionally, the studies demonstrated that with the spread of misinformation, there are consequences and issues. From damaging company reputations to influencing and shaping public opinions and influencing emotional reactions, these findings point to the fact that misinformation spreads most rapidly with these algorithm platforms designed for engagement, added to the lack of digital literacy to evaluate online posts critically.

Together, these documents provide a deeper understanding and answer to our research question by showing how algorithmic design, digital literacy, and user behavior interact to amplify and generate misinformation across social media and video platforms. One study by Guess, Nagler, and Tucker (2019) showed that false news spread faster mainly because people shared posts that felt new or emotional. Additionally, other studies have found that misinformation is primarily shared by certain groups, particularly older adults who struggle with digital literacy. The cycle from algorithms amplifying posts that can trigger strong emotional reactions, users interact and respond more quickly with emotional content, combined with low digital literacy, preventing many people from recognizing the false information they're seeing. The articles analyzed demonstrate that the spread of misinformation is both a technical and a social issue. Overall, the research shows that misinformation is a problem that requires addressing both the design and users' behaviors.

Conclusions and Recommendations

Based on our findings, our research question was answered. However, the answer isn't a single answer. The articles provide clarity in several ways; however, there remain areas for further research to improve understanding. The articles we provided addressed social and technical issues related to misinformation. The roles of algorithms and digital literacies, and how users' behaviour contributes to the spread of fake content. Because each article has more reasons and more angles to view the problem from, it could become more challenging and times to piece together and combine them for one complete explanation. For example, the first article by Aïmeur et al. focused on how platform design increases the spread of emotional content, while Chen and Cheng focused more on how users interpreted believable misinformation and the response to that. One study found that false news spreads faster because people tend to share emotional content. In contrast, another study found that this behaviour is common among older adults with lower digital literacy. These articles gave a deeper dive into showing us how misinformation spreads so quickly, but we could have included more sources that examined specific platforms instead of more general ones.

Overall, our research clearly shows that misinformation is not the result of a single factor, but rather a cycle shaped by technology and human behavior. Engagement based algorithms push content that triggers a more emotional response, while users without digital literacy and knowledge are more likely to share those posts of misinformation. This is even clearer in the research showing that older adults had much more misinformation than younger users, pointing to the fact that digital literacy among different age groups has large gaps. These issues reinforce one another, leading to larger problems, including damaged reputations, reduced trust, and the rapid spread of harmful or false narratives. Our findings show how important it is to address these issues if we want to reduce the spread of misinformation.

Recommended Solutions

Based on the analysis of the collected data, our research indicates that there is no single solution to the spread of misinformation. Instead, effective intervention requires a multi-layered approach that addresses both the technological design of the platform and the social behaviors of users. We recommend the following strategies derived from our Literature Review:

First, social media platforms must implement “algorithmic friction” to counter the speed of dissemination. As proposed by Margolin and Wang (2025), platforms should adopt an “Attestation Framework”. Currently, the cost of sharing false information is near zero. By requiring users to explicitly confirm the accuracy of a post before sharing it (attesting to its truth), platforms can create a psychological pause, a moment of friction, that disrupts the rapid, emotional reaction that leads to viral misinformation. This shifts the focus from simple engagement (clicks and likes) to user accountability.

Second, digital literacy initiatives must be redesigned to be demographic-specific rather than broad-based. Our findings from Guess, Nagler, and Tucker (2019) identified that users over 65 are significantly more likely to share fake news. Therefore, educational interventions should not be generic; they should be specifically targeted at older populations who may be less familiar with the subtle design cues of digital platforms. Resources should focus on helping these users distinguish between legitimate news sources and "novel" or sensationalized content that is designed to exploit cognitive biases.

Finally, organizations and policymakers must shift from reactive to proactive strategies. As suggested by Chen et al. (2023) and Chen and Cheng (2023), relying solely on "message correction" (fact-checking after the fact) is often too slow, as falsehoods travel faster than the truth. Instead, companies should invest in "pre-bunking" strategies—building a reputation for reliability and educating audiences on potential misinformation narratives before they spread. This combination of structural changes (attestation), targeted education (for seniors), and proactive corporate policy offers the most robust path toward mitigating the sociotechnological problem of misinformation.

Recommendations for Further Research

To get a complete overview of our topic of Misinformation in media, we recommend researching more in depth how social media platforms' algorithms work to produce more of certain information. Future studies should look at individual platforms and compare how different recommendation algorithms influence the spread of emotional or misleading information. Additionally, we recommend conducting further research to compare age gaps in digital literacy and to evaluate whether digital literacy training for older adults reduces misinformation sharing over time.

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